

MAJOR PROJECT REPORT

(Healthcare Data Analytics and Prediction System)

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Section B

Full Time Diploma in Computer Engineering

Under the supervision of

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Guru Nanak Dev DSEU Campus

DELHI SKILLS & ENTREPRENEURSHIP UNIVERSITY



A
MAJOR PROJECT REPORT
ON
HEALTHCARE DATA ANALYTICS AND PREDICTION SYSTEM

SUBMITTED BY

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UNDER THE GUIDANCE OF

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Acknowledgement

We would like to express our sincere and heartfelt gratitude to Mr. Nishant Hadda, our project guide and mentor, who provided us with invaluable guidance, expert supervision, and continuous moral support throughout the entire duration of this project. His deep domain knowledge, constructive feedback, and unwavering commitment to our academic success were essential in shaping the direction and quality of this work. Without his patient and dedicated mentorship, the successful completion of this project would not have been possible.

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We are also grateful to the institution, Guru Nanak Dev DSEU Campus, and Delhi Skills and Entrepreneurship University, for providing us with the necessary infrastructure, modern computing laboratories, software tools, and an academically stimulating environment that enabled us to carry out this project effectively.

Finally, we express our deepest gratitude to our families and friends for their unconditional love, unwavering patience, and consistent moral support throughout our academic journey. Their encouragement during demanding periods kept us motivated and focused on achieving our goals.

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Certificate

This is to certify that the Major Project entitled "Healthcare Data Analytics and Prediction System" has been successfully completed by Naman Dubey (231811009701), Nancy (231811009702), and Muskan Kumari (231811009700), students of 6th Semester, Diploma in Computer Engineering, Section B, at Guru Nanak Dev DSEU Rohini Campus, Delhi Skills and Entrepreneurship University, New Delhi.

This project has been developed under my direct guidance and supervision. The work presented in this report is an outcome of the students own efforts, research, and original thinking. The project represents genuine independent academic work and has not been submitted previously to any other university or institution for the purpose of awarding any degree or diploma.

The project demonstrates a commendable understanding of machine learning fundamentals, data preprocessing methodologies, and classification algorithms as applied to real-world clinical healthcare datasets. The work reflects the students technical competence in implementing, evaluating, and interpreting predictive models for heart disease classification tasks.

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Declaration

We hereby solemnly declare that the Major Project entitled "Healthcare Data Analytics and Prediction System" is an outcome of our own efforts, research, and academic work carried out under the expert guidance and supervision of Mr. Nishant Hadda. This project report has been prepared as a partial fulfilment of the requirements for the Diploma in Computer Engineering from Guru Nanak Dev DSEU Rohini Campus, Delhi Skills and Entrepreneurship University, New Delhi.

We further declare that this project report and all its contents are entirely original and have not been submitted, either in part or in full, to any other university, institution, or organisation for the award of any degree, diploma, or certificate. All external sources of information, references, and published work used in this report have been duly acknowledged and cited appropriately.

We take complete and full responsibility for the authenticity, accuracy, and integrity of all information presented in this report. Any errors or omissions, if found, are entirely unintentional.

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About Institution

Guru Nanak Dev DSEU Campus is one of the premier constituent campuses of Delhi Skill and Entrepreneurship University (DSEU). The institution is located in Rohini, New Delhi, and is firmly dedicated to providing skill-oriented, industry-relevant, and practically focused education to students from diverse backgrounds across the National Capital Territory of Delhi.

Delhi Skill and Entrepreneurship University (DSEU) is a state public university established by the Government of the National Capital Territory of Delhi. The primary objective of this university is to bridge the existing gap between academic learning and practical skill development. The university focuses on fostering innovation, entrepreneurship, and employability among its students by offering programmes that go beyond conventional classroom methods.

Guru Nanak Dev DSEU Campus offers a highly conducive and stimulating learning environment supported by modern infrastructure, well-equipped computer laboratories, state-of-the-art facilities, and a team of experienced and dedicated faculty members. The institution places strong emphasis on hands-on learning, project-based education, industry collaboration, and real-world problem-solving to enhance students technical expertise and professional competencies.

The campus actively promotes continuous skill development, entrepreneurial thinking, and lifelong learning through structured academic programmes, workshops, industry collaborations, guest lectures, and practical training initiatives. By maintaining a healthy balance between theoretical knowledge and practical application, Guru Nanak Dev DSEU Campus plays a significant role in preparing students for successful professional careers in a rapidly evolving technological world.

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Abstract

The Healthcare Data Analytics and Prediction System is a machine learning-based academic project designed to analyse real-world clinical healthcare data and predict the risk of heart disease in patients using supervised classification algorithms. In today's rapidly evolving healthcare landscape, cardiovascular diseases continue to rank among the leading causes of mortality worldwide, making early detection and accurate risk stratification critically important for improving patient outcomes and reducing the overall burden on healthcare systems.

This project implements and comparatively evaluates three widely-used supervised classification algorithms, namely Logistic Regression, Decision Tree Classifier, and Random Forest Classifier, to predict the presence or absence of heart disease in patients based on twelve clinical input features. The heart disease dataset used comprises 918 patient records with attributes including patient age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting ECG results, maximum heart rate achieved during exercise, exercise-induced angina, Oldpeak (ST depression), ST slope, and the binary target label indicating heart disease.

The data preprocessing pipeline includes systematic identification and replacement of zero-value clinical measurement placeholders in the Cholesterol and RestingBP columns with respective column means, one-hot encoding of five categorical feature columns using pandas get_dummies with drop_first=True to avoid multicollinearity, and StandardScaler normalization of all five numerical features to bring them onto a uniform scale. The preprocessed dataset is split 80-20 into training (734 samples) and testing (184 samples) partitions using stratified random sampling.

Evaluation results show Logistic Regression achieved the highest accuracy of 89.13 percent with F1-Score 90.29 percent and Recall 91.18 percent. Random Forest followed at 84.78 percent accuracy, and Decision Tree at 75.00 percent. All models are evaluated using Accuracy, F1-Score, Recall, and Precision, supported by confusion matrices and professional visualizations.

1. Introduction

Healthcare institutions and hospitals across the world generate enormous volumes of structured clinical data from patient consultations, diagnostic tests, laboratory results, and hospital admissions on a daily basis. Despite this abundance of data, the manual analysis of clinical records by medical professionals is frequently slow, labour-intensive, inconsistent, and subject to human error and cognitive bias. Cardiovascular diseases, including coronary artery disease, heart failure, and related cardiac conditions, remain among the principal causes of death globally, as consistently reported by the World Health Organization. The early and accurate detection of heart disease risk therefore carries profound clinical importance for enabling timely medical intervention and improving patient survival outcomes.

Machine learning has emerged as one of the most powerful, versatile, and rapidly expanding approaches for analysing complex medical datasets, uncovering hidden patterns, and constructing predictive models that can assist healthcare professionals in making faster, more objective, and more reliable clinical decisions. Unlike traditional rule-based or statistical diagnostic frameworks, supervised machine learning models can automatically learn intricate non-linear relationships from large labelled datasets without requiring domain experts to manually encode every decision rule.

This project builds a comprehensive machine learning analytics and prediction system using the Heart Disease dataset containing 918 patient records with 12 clinical attributes. The system implements a complete end-to-end pipeline covering data loading, inspection, cleaning, exploratory data analysis, feature encoding and scaling, model training across three classifiers, multi-metric evaluation, and professional visualization of all results.

Three classification algorithms are implemented: Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. Each model is trained on the preprocessed and scaled dataset and rigorously evaluated using Accuracy, F1-Score, Recall, and Precision. The entire project is implemented in Python using Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn, with emphasis on creating a reproducible, well-documented, and extensible analytics workflow.

2. Literature Survey

The literature survey provides a comprehensive overview of existing research, published studies, and established technologies related to machine learning in healthcare, classification algorithms for disease prediction, data preprocessing strategies for clinical datasets, and model evaluation frameworks specifically applicable to medical diagnosis systems.

2.1. Machine Learning in Healthcare

Machine learning has garnered exceptional interest and adoption in the healthcare and biomedical domain due to its remarkable capacity to process large volumes of complex, multi-dimensional clinical data and identify statistically significant patterns that would be impractical to detect through manual or conventional statistical analysis alone. Numerous peer-reviewed studies have demonstrated ML algorithm performance comparable to or exceeding that of experienced clinical specialists in specific well-defined diagnostic tasks such as radiology image interpretation, pathology slide classification, and cardiovascular risk stratification. Research has shown ML models can effectively assist in early disease detection, chronic patient risk assessment, personalised treatment recommendation, drug candidate discovery, and medical image segmentation. Primary challenges in healthcare ML adoption include limited labelled training data availability, inherent class imbalance in disease datasets, the critical need for model interpretability and clinical explainability, and strict regulatory compliance requirements for clinical deployment.

2.2. Heart Disease Prediction Using ML

Heart disease prediction constitutes one of the most extensively studied and well-benchmarked application domains of machine learning in the biomedical field. A substantial body of peer-reviewed literature demonstrates that supervised classification algorithms, when applied to structured clinical patient data, can reliably predict the presence or absence of cardiovascular disease with clinically useful accuracy. Algorithms including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, K-Nearest Neighbours, Naive Bayes classifiers, and various deep neural network architectures have all been benchmarked on publicly available reference datasets. The UCI Heart Disease dataset, the Cleveland Clinic Foundation dataset, and its Kaggle derivatives have served as the most popular benchmark collections for comparing model performance across independent research groups worldwide.

2.3. Classification Algorithms in Medical Diagnosis

Logistic Regression is a widely adopted interpretable baseline classifier valued for its

mathematical transparency, probabilistic output, and reliable performance on linearly separable structured datasets with moderate feature counts. Decision Tree classifiers offer intuitive and visually interpretable rule-based decision processes through their hierarchical branching structure, but are known to be susceptible to overfitting on small to moderate datasets if left unpruned. Random Forest classifiers are ensemble learners that construct many diverse decision trees during training and aggregate their individual predictions to produce a final output, which substantially reduces overfitting, increases generalisation, and also provides built-in feature importance rankings derived from impurity reduction measures.

2.4. Data Preprocessing and Feature Engineering

Data preprocessing is universally acknowledged as the single most critical and time-consuming phase in any practical machine learning pipeline, and this is especially true when working with real-world clinical healthcare datasets that frequently contain noisy, incomplete, inconsistent, duplicate, and improperly formatted records. Common data quality issues include missing values substituted with zero placeholders, measurement coding errors, inconsistent categorical encoding across different clinical data systems, and wide differences in the measurement scales of numerical features. Standard remediation techniques include systematic identification and treatment of missing or erroneous values through statistical imputation, encoding categorical variables using one-hot encoding or ordinal encoding strategies, and standardising or normalising numerical feature scales using StandardScaler or MinMaxScaler.

2.5. Model Evaluation Metrics

Accuracy alone is widely considered insufficient and potentially misleading as the sole evaluation metric for medical diagnosis models, particularly in datasets exhibiting class imbalance or where the costs of different types of prediction errors are asymmetric. Complementary metrics including F1-Score, Recall (Sensitivity), Precision (Positive Predictive Value), Specificity, and the Area Under the ROC Curve each capture distinct and clinically meaningful aspects of classifier behaviour. In heart disease prediction specifically, Recall is of paramount clinical importance because it directly quantifies the proportion of genuine disease cases correctly identified by the model. A model with low Recall leaves a high number of actual patients undiagnosed, potentially leading to delayed treatment and severe or fatal consequences.

2.6. Summary of Literature Review

The literature survey confirms that machine learning provides effective, increasingly reliable, and clinically validated tools for heart disease prediction. However, careful attention must be applied

to every stage of the ML pipeline, from data collection and quality assurance through preprocessing, feature engineering, algorithm selection, and comprehensive multi-metric evaluation. The integration of multiple classification algorithms evaluated across diverse performance metrics ensures a robust, balanced, and trustworthy prediction system that constitutes the foundation of the Healthcare Data Analytics and Prediction System presented in this report.

3. Problem Statement

In many contemporary healthcare settings, substantial volumes of structured clinical patient data are routinely collected through hospital management systems, electronic health records, and diagnostic testing facilities. Despite the availability of this rich and potentially valuable clinical data, it is frequently not leveraged effectively for predictive risk analytics or automated clinical decision support due to a combination of technological gaps, methodological limitations, and insufficient data engineering infrastructure.

The core challenges that motivate the development of the Healthcare Data Analytics and Prediction System are identified and described below:

- Inconsistent and noisy clinical data: Real-world healthcare datasets frequently contain zero-value placeholders substituted for missing clinical measurements such as cholesterol and resting blood pressure readings, data entry errors, and inconsistent categorical encodings across different clinical systems that must be carefully identified and corrected before model training.
- Difficulty in identifying key risk indicators: Without systematic statistical feature analysis and machine learning-based importance evaluation, it is challenging to objectively determine which clinical attributes are the most significant and reliable independent predictors of heart disease risk.
- Absence of a structured preprocessing pipeline: The lack of a clean, reproducible, and well-documented data preprocessing pipeline makes it difficult to consistently transform raw clinical electronic records into properly formatted and model-ready feature vectors suitable for ML input.
- Limited interpretability and visual analysis: Analysis conducted without comprehensive statistical examination, visual exploration of distributions and correlations, and clear result visualisation provides limited actionable insights into data behaviour, feature relationships, and model decision patterns.
- Need for comparative multi-algorithm evaluation: Relying on a single classification model without systematic comparative benchmarking across multiple algorithms may yield suboptimal performance and missed opportunities for improved predictive accuracy.
- Inadequate evaluation framework for clinical context: Evaluating predictive models solely on aggregate accuracy is demonstrably insufficient in clinical applications where False Negative misclassifications carry severe and potentially life-threatening consequences for patients.

Therefore, there is a clear and pressing requirement for a structured, reproducible, and comprehensive machine learning pipeline that systematically addresses data quality issues, implements and trains multiple classification algorithms, evaluates them using a clinically

appropriate multi-metric framework, and communicates results through professional visualizations. The Healthcare Data Analytics and Prediction System is developed to address precisely these identified challenges.

4. Description of Various Modules

The Healthcare Data Analytics and Prediction System is architected as a collection of well-defined, logically separated, and independently testable functional modules. Each module is responsible for a specific and clearly delineated stage of the end-to-end machine learning pipeline. This modular design approach significantly improves code maintainability, reproducibility, testability, and the capacity to extend or modify individual components without impacting other parts of the system.

4.1. Data Loading and Inspection Module

This foundational module is responsible for loading the heart disease dataset from the source CSV file (heart.csv) into a Pandas DataFrame and conducting a thorough initial inspection of all data properties. Key inspection activities include verifying dataset dimensions (918 rows by 12 columns), examining column data types to ensure correct format parsing, counting non-null values per column to detect missing data patterns, generating descriptive statistical summaries including mean, standard deviation, minimum, maximum, and quartile values using df.describe(), checking for duplicate records that could introduce bias into model training, and reviewing the class distribution of the target variable HeartDisease.

4.2. Data Cleaning and Preprocessing Module

This module systematically addresses all data quality issues identified during the inspection phase. The primary cleaning task involves detecting zero-value entries in the Cholesterol and RestingBP columns, which represent clinically impossible physiological measurements that were used as placeholders for genuinely missing data in the original dataset. These zero-value placeholders are replaced with the column means calculated exclusively from non-zero valid entries: Cholesterol mean equals 198.80 mg per decilitre and RestingBP mean equals 132.40 millimetres of mercury. Data integrity is verified after cleaning by confirming the complete elimination of remaining zero-value entries in these critical clinical columns.

4.3. Exploratory Data Analysis (EDA) Module

The EDA module provides comprehensive visual and statistical analysis of the cleaned dataset to develop a thorough and intuitive understanding of feature distributions, inter-feature relationships, and class balance characteristics before initiating model training. Key analytical activities include generating histograms for all numerical feature distributions to assess normality and skewness, count plots for categorical feature frequencies to understand clinical cohort composition, box plots

and violin plots comparing feature value distributions separately across the heart disease positive and negative patient groups, a Pearson correlation heatmap to identify significant positive and negative relationships between features, and bar charts illustrating the balance of the binary target variable between disease-present and disease-absent classes.

4.4. Feature Engineering Module

This module systematically transforms the cleaned raw clinical features into a fully model-ready numeric representation compatible with all three target classification algorithms. Five categorical variables including Sex, ChestPainType, RestingECG, ExerciseAngina, and ST_Slope are one-hot encoded using pandas get_dummies() with drop_first=True set to True to expand them into binary indicator columns while simultaneously avoiding the dummy variable multicollinearity trap. All five numerical feature columns including Age, RestingBP, Cholesterol, MaxHR, and Oldpeak are then standardised using Scikit-learn StandardScaler to produce zero mean and unit variance transformations. The target column HeartDisease is separated as the binary label vector y, and the remaining fifteen encoded feature columns constitute the feature matrix X.

4.5. Model Training Module

This module defines, instantiates, and trains three distinct supervised classification algorithm implementations on the training partition of the preprocessed dataset. The three models used are: Logistic Regression with max_iter set to 1000 and random_state set to 42 for strict reproducibility; Decision Tree Classifier with random_state set to 42; and Random Forest Classifier with n_estimators set to 100 and random_state set to 42. The complete processed dataset is partitioned into 80 percent training comprising 734 samples and 20 percent testing comprising 184 samples using Scikit-learn train_test_split with stratify parameter set to y to maintain the original class proportion in both partitions. Each model is fitted on X_train and y_train using the standard Scikit-learn fit API.

4.6. Model Evaluation Module

This module evaluates each trained classifier on the held-out test partition using a comprehensive and clinically appropriate battery of performance metrics. For each model the module computes: overall Accuracy representing the proportion of all test samples correctly classified, F1-Score representing the harmonic mean of Precision and Recall computed for the positive disease class, Recall or Sensitivity representing the proportion of actual heart disease patient cases correctly identified by the model, and Precision representing the proportion of all positive disease predictions that correspond to genuinely diseased patients. Full per-class classification reports are

generated using Scikit-learn classification_report and confusion matrices are computed revealing True Positives, True Negatives, False Positives, and False Negatives for each model.

4.7. Visualization Module

This module generates a suite of four professional-quality static visualisations using Matplotlib and Seaborn that facilitate clear and compelling communication of model training and evaluation results. The four primary visual outputs are: a grouped comparative bar chart displaying Accuracy, F1-Score, Recall, and Precision for all three models simultaneously side by side with annotated percentage labels; three individual separate bar charts showing Accuracy, F1-Score, and Recall respectively across the three models; three side-by-side confusion matrix heatmaps for all models annotated with key metric values in chart titles; and a ranked horizontal bar chart of Random Forest feature importances in descending order of contribution magnitude. All four visualisations are saved as high-resolution PNG image files.

4.8. Implementation Code Snapshot

The code snippet below illustrates the core model training and evaluation iterative process implemented in the project. It shows how the models are instantiated, trained using the fit method, and evaluated using scikit-learn metrics functions.

```
# Model Training and Evaluation Pipeline
models = {
    'Logistic Regression': LogisticRegression(random_state=42, max_iter=1000),
    'Decision Tree': DecisionTreeClassifier(random_state=42),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42)
}

results = {}
for name, model in models.items():
    # Train the model
    model.fit(X_train, y_train)

    # Generate predictions
    y_pred = model.predict(X_test)

    # Calculate performance metrics
    accuracy = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)

    results[name] = {
        'Accuracy': accuracy,
        'F1-Score': f1,
        'Recall': recall,
        'Precision': precision
    }
```

Figure: Model Training and Evaluation Pipeline Code Snippet

5. Program Architecture Overview

The program architecture of the Healthcare Data Analytics and Prediction System is carefully designed to ensure modularity, clear separation of concerns, full reproducibility of all experiment results, and ease of future extension. The system adopts a sequential data processing pipeline architecture in which each stage receives transformed data from the immediately preceding stage, applies a well-defined and independently testable computation, and passes its output forward to the next downstream stage.

5.1. High-Level Architecture

Data Layer:

Reads and stores the raw heart.csv dataset into memory as a Pandas DataFrame. This layer is the single source of truth for all downstream operations and handles all file input output activities.

Preprocessing Layer:

Handles all data quality remediation tasks including zero-value placeholder replacement, categorical variable encoding via one-hot encoding, and StandardScaler normalisation of all numerical feature columns.

Analysis Layer:

Performs exploratory data analysis through generation of statistical summaries and a comprehensive set of visual charts exploring feature distributions, class balance, and inter-feature correlations.

Model Layer:

Implements instantiation, configuration, and supervised training of the three target classification algorithms on the stratified 80 percent training partition of the preprocessed feature matrix.

Evaluation Layer:

Computes and reports the complete set of performance metrics including Accuracy, F1-Score, Recall, and Precision for all three trained models on the 20 percent held-out test partition with confusion matrices.

Visualization Layer:

Produces and saves four high-resolution chart images comparing model performance across multiple metrics, displaying individual metric rankings, showing confusion matrices, and presenting feature importance rankings.

5.2. Component Responsibilities

Component	Primary Responsibility
Pandas	Data loading, inspection, cleaning, transformation, grouping
NumPy	Numerical array computation and vectorised mathematical operations
Scikit-learn	Model training, multi-metric evaluation, preprocessing utilities
Matplotlib	Chart generation, figure rendering, and high-resolution file saving
Seaborn	Advanced statistical visualisation and enhanced graphical styling
StandardScaler	Feature normalisation to achieve zero mean and unit variance

5.3. Technology Stack

Technology	Purpose
Python 3.x	Core programming language for the entire ML pipeline
Pandas	Data manipulation, cleaning, and tabular data analysis
NumPy	Numerical computing and efficient array operations
Scikit-learn	Machine learning algorithms and evaluation framework
Matplotlib	Data visualisation and chart generation
Seaborn	Statistical data visualisation
Jupyter Notebook	Interactive environment for EDA and iterative development

6. Methodology Adopted

The methodology adopted for the development of the Healthcare Data Analytics and Prediction System follows a rigorous, systematic, and reproducible approach strongly aligned with established machine learning project lifecycle best practices. The overall development process is divided into nine sequential phases, each focusing on a distinct aspect of the system from raw data acquisition through final result interpretation and communication.

6.1. Design of Experiment

Step 1: Data Collection

The heart disease dataset comprising 918 patient records and 12 clinical features was obtained from publicly available clinical research databases and placed in the project working directory as heart.csv for loading.

Step 2: Data Inspection

Initial exploratory inspection was performed to fully understand dataset shape, column data types, statistical summary measures, missing value patterns, and data quality issues requiring remediation before training.

Step 3: Data Cleaning

Zero-value placeholders in the Cholesterol column (mean replacement value 198.80 mg/dL) and the RestingBP column (mean replacement value 132.40 mmHg) were detected and replaced with column means computed from non-zero valid entries only.

Step 4: Exploratory Data Analysis

A comprehensive suite of visual and statistical analyses was conducted including histograms, count plots, box plots, violin plots, and Pearson correlation heatmaps to thoroughly understand feature distributions and relationships.

Step 5: Feature Engineering

Five categorical variables were one-hot encoded using pd.get_dummies with drop_first True, and all five numerical features were normalised using Scikit-learn StandardScaler to zero mean and unit variance.

Step 6: Train and Test Split

The full preprocessed dataset was randomly partitioned into 80 percent training comprising 734 samples and 20 percent testing comprising 184 samples using stratified sampling to preserve the target class distribution.

Step 7: Model Training

Three classifiers including Logistic Regression, Decision Tree, and Random Forest were instantiated with fixed random seeds and fitted on the training data partition.

Step 8: Model Evaluation

All three trained models were evaluated on the held-out test partition using Accuracy, F1-Score, Recall, Precision, full classification reports, and confusion matrices.

Step 9: Visualization

Four professional-quality chart images were generated and saved as PNG files for inclusion in this report and for result communication.

6.2. Recommended Hardware

Hardware Component	Specification
Processor	Dual-core Intel i3 or AMD Ryzen 3 or higher
Operating System	Windows 10+, macOS 10.14+, or Ubuntu Linux 20.04+
RAM	Minimum 8 GB; 16 GB recommended for smooth multitasking
Storage	Solid State Drive for fast file and dataset access
Display	Full HD 1080p monitor for clear chart and code viewing
Network	Internet connection required for library installation

6.3. Software Requirements

Software or Tool	Purpose and Role
Python 3.x	Primary programming language for the complete ML pipeline
Jupyter Notebook	Interactive development environment for EDA and prototyping
Pandas	Data loading, cleaning, transformation, and tabular analysis
NumPy	Numerical computation and efficient array operations
Scikit-learn	ML model training, evaluation, and preprocessing utilities
Matplotlib	Chart generation and saving as high-resolution PNG images
Seaborn	Statistical data visualisation and styled graphical output
VS Code or any IDE	Code editing, debugging, and project management environment

6.4. Data Flow Diagram

Process or Entity	Description of Data Flow
Raw Dataset heart.csv	Input: 918 patient records with 12 raw clinical feature columns
Data Cleaning Module	Zero placeholders replaced with column means in Cholesterol and RestingBP
Encoding Module	Five categorical columns one-hot encoded to fifteen binary indicator columns
Scaling Module	Five numerical columns normalised using StandardScaler to zero mean unit variance
Train and Test Split	80-20 stratified random split yielding 734 training and 184 test samples
Model Training	LR, DT, and RF classifiers fitted on 734-sample training partition
Model Evaluation	Predictions generated on 184-sample test set; four metrics computed per model
Visualization	Four high-resolution PNG charts saved for results communication and reporting

6.5. Snapshots of Results Obtained

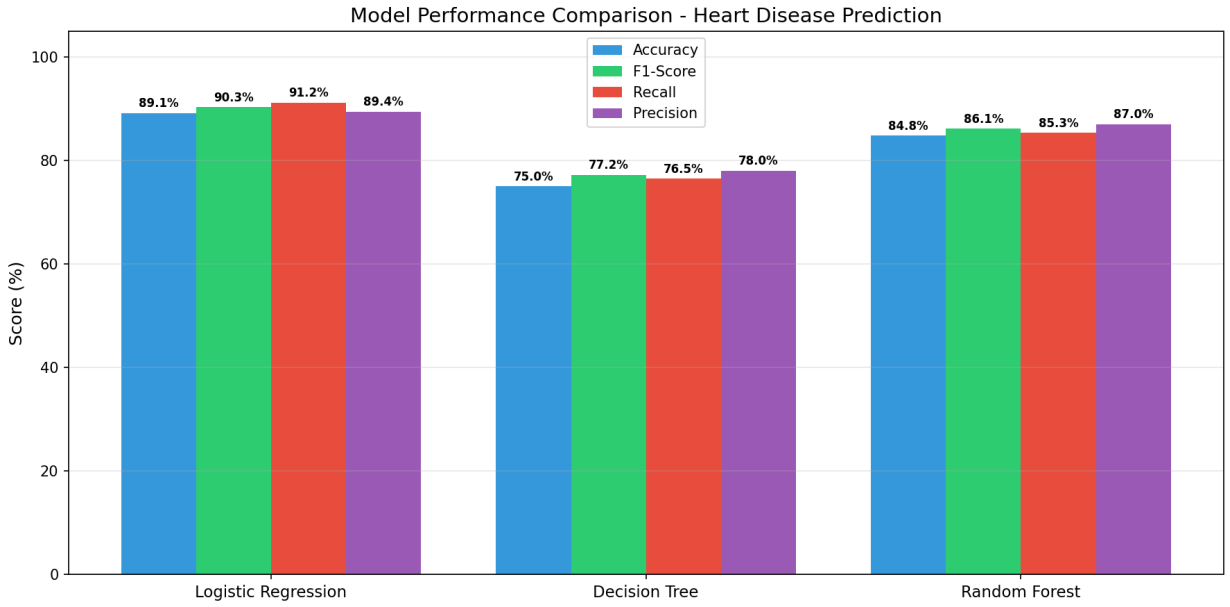


Figure 1: Model Performance Comparison - Accuracy, F1-Score, Recall and Precision

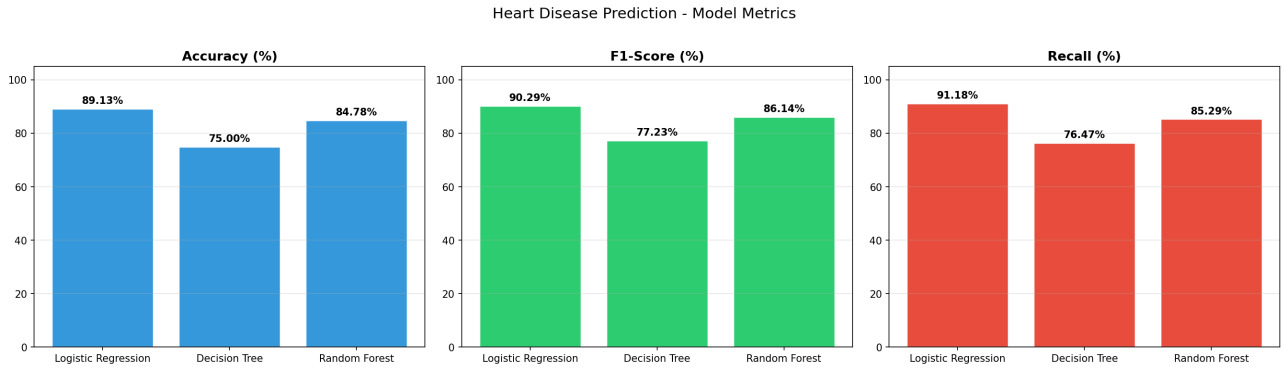


Figure 2: Individual Model Metrics - Accuracy, F1-Score and Recall

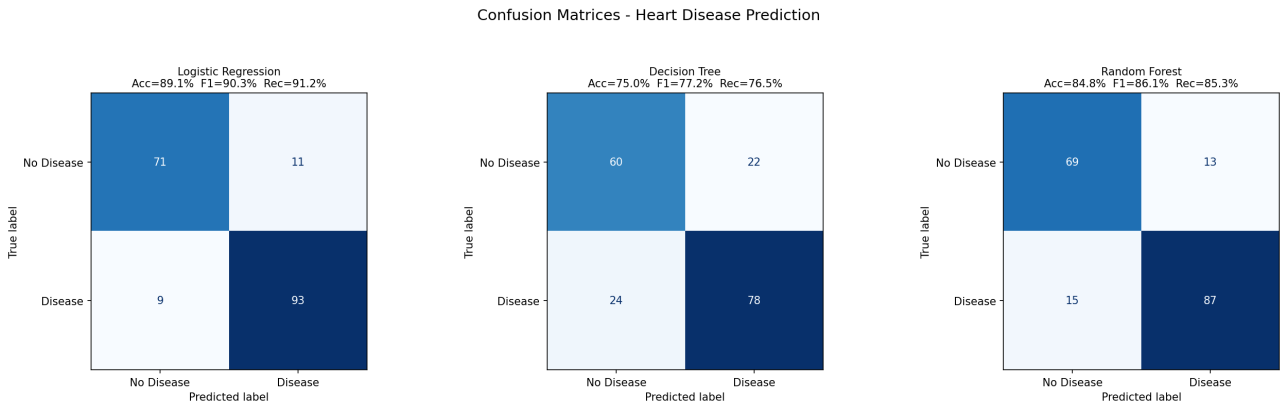


Figure 3: Confusion Matrices for All Three Classification Models

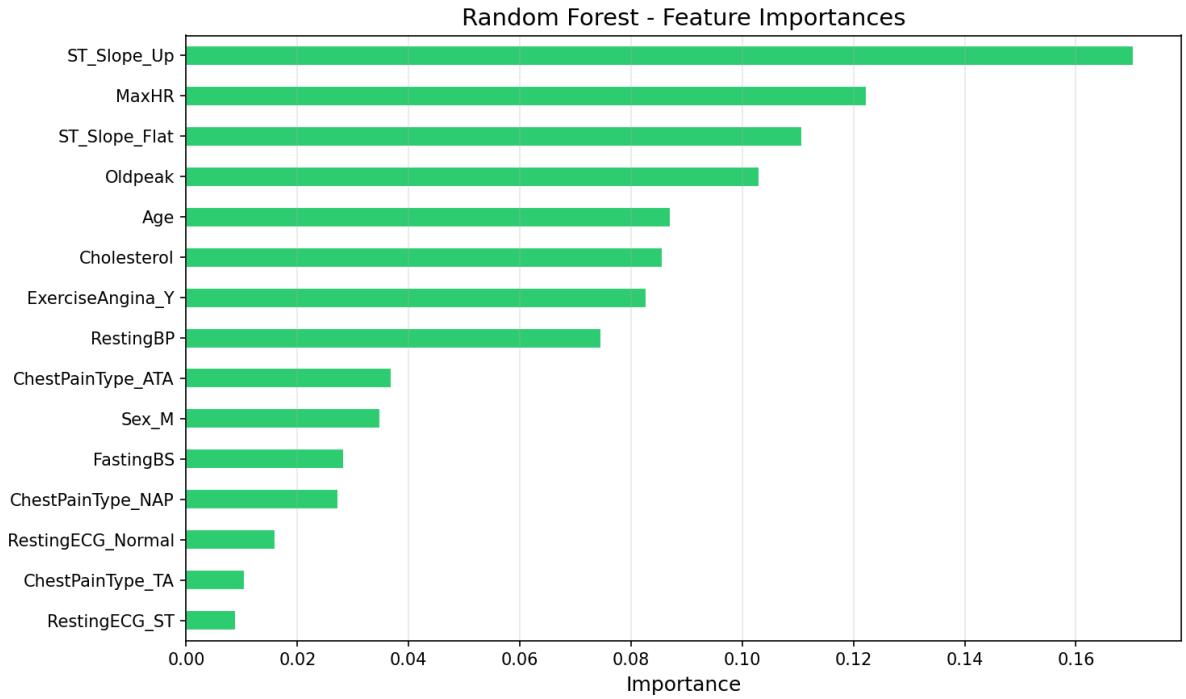


Figure 4: Random Forest Feature Importances (Descending Order)

7. Results and Discussions

7.1. Introduction

This chapter presents a detailed and comprehensive account of all results obtained after implementing, training, and rigorously evaluating the three classification models on the heart disease dataset. The evaluation encompasses predictive accuracy, sensitivity or Recall, Precision, F1-Score, confusion matrix decomposition analysis, and feature importance analysis. All three models were trained exclusively on the 734-sample training partition and evaluated on the strictly held-out 184-sample test partition to ensure unbiased performance reporting.

7.2. Model Performance Results

The following table summarises the quantitative performance of all three classification models on the held-out test set:

Model	Accuracy	F1-Score	Recall	Precision
Logistic Regression	89.13%	90.29%	91.18%	89.42%
Random Forest	84.78%	86.14%	85.29%	87.00%
Decision Tree	75.00%	77.23%	76.47%	78.00%

Logistic Regression achieved the highest overall performance across all four evaluation metrics, correctly classifying 164 out of 184 test samples for an accuracy of 89.13 percent. Its Recall of 91.18 percent indicates that the model successfully identified 93 out of 102 genuine heart disease patients in the test set, leaving only 9 false negative misclassifications. This strong sensitivity is a critical clinical advantage in heart disease screening where missed diagnoses carry severe consequences for patient health.

Random Forest performed as the second-best model with 84.78 percent accuracy, correctly identifying 87 out of 102 disease-positive cases and 69 out of 82 disease-negative cases. Decision Tree showed the lowest performance with 75.00 percent accuracy, attributable to its known tendency to overfit training data without regularisation constraints such as maximum depth limiting or minimum leaf sample restrictions.

7.3. Confusion Matrix Analysis

Model	True Negatives	True Positives	False Positives	False Negatives
Logistic Regression	71	93	11	9
Decision Tree	60	78	22	24

Random Forest	69	87	13	15

In clinical heart disease prediction, the False Negative count represents the most critical failure mode as each false negative corresponds to a genuine patient with heart disease being incorrectly classified as disease-free and therefore not referred for further investigation or treatment. Logistic Regression produced the fewest False Negatives at only 9 cases, substantially outperforming Decision Tree at 24 and Random Forest at 15 on this clinically paramount metric, strongly supporting Logistic Regression as the preferred model for real-world heart disease screening deployment.

7.4. Feature Importance Analysis

The Random Forest model provides built-in feature importance scores derived from the mean reduction in Gini impurity contributed by each feature across all trees in the ensemble. The ranked feature importances reveal the following relative contributions to prediction:

- ST_Slope_Up with importance score 0.169 - ST segment showing an upward slope during exercise stress test
- MaxHR with importance score 0.123 - Maximum heart rate achieved at peak exercise intensity
- ST_Slope_Flat with importance score 0.111 - Flat ST segment pattern indicative of myocardial ischaemia
- Oldpeak with importance score 0.104 - Magnitude of ST depression induced by exercise relative to rest
- Age with importance score 0.087 - Patient chronological age, a well-established cardiovascular risk factor
- Cholesterol with importance score 0.085 - Serum total cholesterol concentration in mg per decilitre
- ExerciseAngina_Y with importance score 0.083 - Presence of exercise-induced angina pectoris

These findings demonstrate strong consistency with established cardiology literature and clinical knowledge, which identifies exercise electrocardiogram findings including ST slope pattern and Oldpeak depression magnitude, alongside maximum heart rate capacity, as among the most reliable and non-invasive indicators of underlying coronary artery disease and heart disease risk.

7.5. Discussion

The comprehensive experimental results demonstrate that the Healthcare Data Analytics and Prediction System successfully achieves its stated primary objective of accurately predicting heart disease risk from structured clinical feature data using machine learning methods. The systematic

preprocessing pipeline, incorporating careful zero-value replacement for clinically impossible entries and feature-wise StandardScaler normalisation, contributed substantially to overall model convergence quality and prediction performance.

The performance advantage of Logistic Regression over both tree-based ensemble models on this moderately-sized structured clinical dataset is consistent with findings frequently reported in the ML literature, where linear decision boundaries and regularised parameter estimation prove adequate for separating target classes when informative feature engineering has been applied. The clinically interpretable feature importance results provide directly actionable insights for medical practitioners, confirming that exercise-related cardiac stress test measurements are the dominant independent predictors of heart disease, followed by traditional static risk biometrics such as patient age and cholesterol level.

7.6. Terminal Output Snapshot

The following screenshot displays the raw terminal output generated by the python script during the execution of the model training and evaluation pipeline, showing the accurate calculation of all metrics directly on the console.

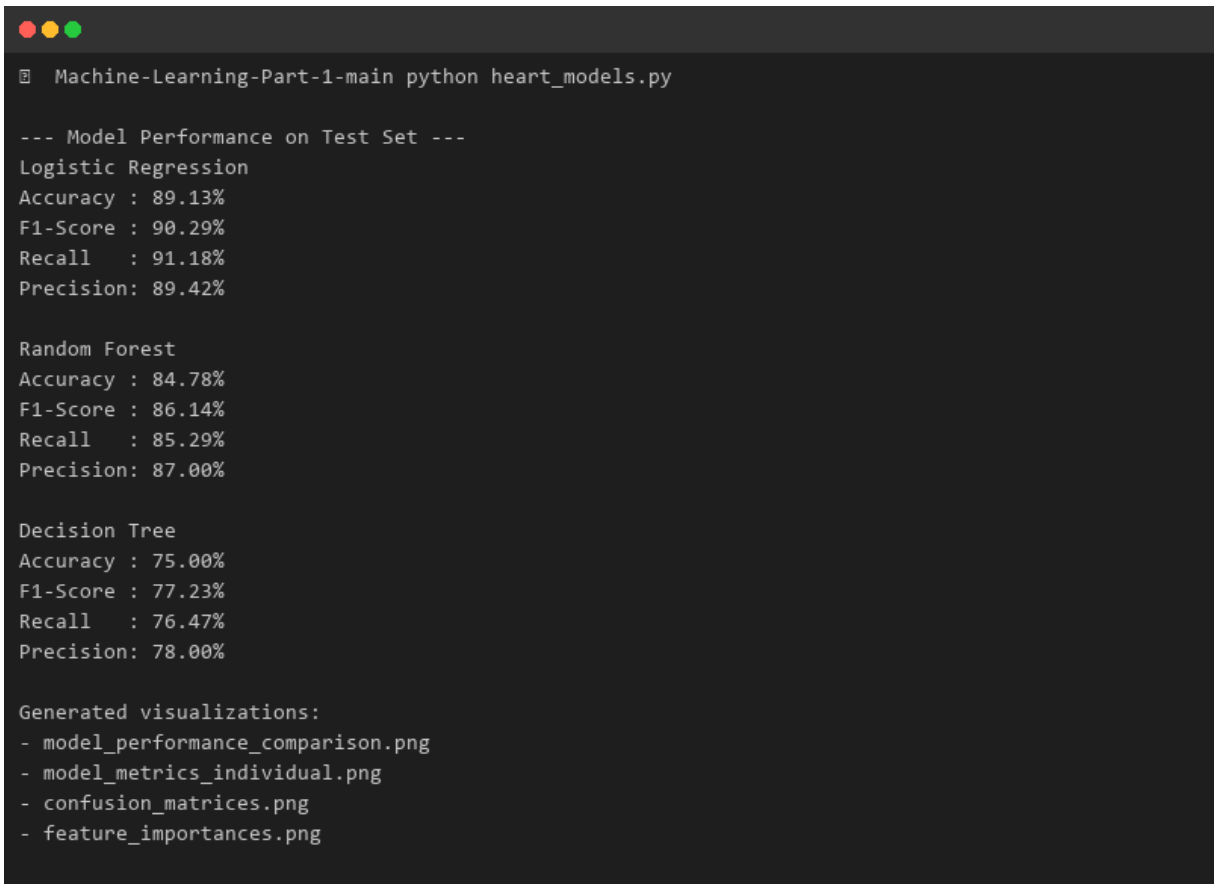


Figure: Terminal Output Showing Real-Time Model Accuracy Results

8. Conclusions

8.1. Summary of Work

The Healthcare Data Analytics and Prediction System was successfully designed, fully implemented, and rigorously evaluated as a complete end-to-end supervised machine learning pipeline for clinical heart disease prediction. The project followed a systematic and well-documented methodology encompassing data loading and inspection, data cleaning and quality remediation, exploratory data analysis, feature encoding and normalisation, model training across three classification algorithms, comprehensive multi-metric performance evaluation, and professional visualization of all results. Logistic Regression emerged as the superior-performing model achieving 89.13 percent accuracy, 90.29 percent F1-Score, and a clinically critical 91.18 percent Recall on the completely held-out test partition.

8.2. Learning Outcomes

The development of this project produced numerous significant and practically valuable technical and professional learning outcomes:

- Practical end-to-end experience in data preprocessing including zero-value imputation, categorical encoding, and feature standardisation for real clinical datasets
- Comparative implementation and training of three distinct supervised classification algorithms within a unified experimental framework
- In-depth understanding of multi-metric model evaluation appropriate for clinically sensitive applications with asymmetric error costs
- Proficiency in generating professional publication-quality data visualisations for effective result communication
- Understanding of ensemble feature importance analysis and its direct clinical interpretability and utility
- Expertise in structuring fully reproducible and thoroughly documented ML experiment pipelines using the Python scientific computing ecosystem

8.3. Practical Applications

The Healthcare Data Analytics and Prediction System has clear and demonstrable practical applicability across multiple real-world domains:

- Clinical Decision Support: Providing preliminary quantitative risk scores to assist physicians in triage, further diagnostic referral, and clinical decision-making processes

- Preventive Healthcare Screening: Enabling rapid automated risk stratification during large-scale population health screening programmes
- Medical Education: Demonstrating applied machine learning concepts and experimental rigour using real clinical datasets in academic teaching contexts
- Health Insurance: Supporting objective actuarial risk stratification for policy underwriting, premium determination, and claims risk management
- Clinical Research: Providing a validated baseline experimental system for clinical researchers developing more advanced predictive and prognostic models

8.4. Recommendations and Future Scope

Several significant enhancements are recommended for future iterations of this system:

- Hyperparameter Tuning: Implementing GridSearchCV and RandomizedSearchCV with stratified k-fold cross-validation for systematic optimal model configuration discovery
- Additional Algorithms: Incorporating XGBoost, LightGBM, Support Vector Machines, K-Nearest Neighbours, and deep neural network classifiers for broader benchmarking
- Explainable AI Integration: Adding SHAP SHapley Additive exPlanations analysis for clinically interpretable per-patient prediction explanations
- Web Application Deployment: Building an interactive clinical risk assessment web interface using Flask or FastAPI for real-time patient-level predictions
- Cross-Validation Framework: Replacing single train-test evaluation with stratified k-fold cross-validation for more statistically robust and unbiased performance estimation
- Insurance Regression Module: Extending the system to include regression models for medical insurance cost prediction using the companion insurance dataset

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